SIMATS ENGINEERING ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0510

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3) Assessment Tool Weightage: 10%

Code of Conduct:

I P. Jyothika/ 192311156 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Jyothika

Rubrics for Evaluation

Criteria	*Weightage*	*Level 4 – Exemplary*	*Level 3 – Proficient*	*Level 2 – Developing*	*Level 1 – Beginning*	*Marks(100)*
Understanding of Concepts	*25%*	*Demonstrates thorough, accurate understanding of concepts*	*Good understanding with minor gaps*	*Basic understanding with some misconceptions*	*Poor understanding; major misconceptions*	
Explanation	*25%*	*Detailed, wellexplained answers with relevant examples*	*Clear explanation with adequate details*	*Limited explanation; lacks depth*	*Poor or unclear explanation*	
Application	*20%*	*Effectively applies concepts with strong analytical ability*	*Applies concepts with minor errors*	*Limited application with weak analysis*	*No application or incorrect analysis*	
Organization	*15%*	*Well-structured answers with logical flow and coherence*	*Organized with minor issues in flow*	*Basic structure, lacks clarity*	*Poor organization, incoherent answers*	

Accuracy	*10%*	*Completely accurate and covers all required points*	*Mostly accurate with minor omissions*	*Partially correct with missing points*	*Incorrect answers with major gaps*	
Clarity	*5%*	*Clear, neat, professional presentation*	*Understandabl e with minor issues*	*Limited clarity, readability issues*	*Poorly written, difficult to understand*	

1.An image appears too bright.

Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

Answer

When an image appears too bright, most of the pixel intensity values are concentrated near the higher intensity range (close to 255). As a result, important details in bright regions become difficult to observe.

The most suitable grey level transformation is the Negative Transformation.

Concept:

Negative transformation converts each pixel intensity using:

$$s = 255 - r$$

where **r** is the original pixel value and **s** is the transformed value.

Application:

The transformation inverts the brightness of the image. Bright pixels become dark and dark pixels become bright, making hidden details more visible.

Justification:

This method is especially useful for medical images and X-ray images where bright regions hide important information.

Expected Outcome:

The transformed image provides improved visibility and better contrast, making image details easier to analyze.

2.A dark image lacks detail.

Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning. **Answer**

Introduction

A dark image contains mostly low-intensity pixel values, making objects and important details difficult to identify. This may occur due to poor lighting or underexposure. To improve visibility, an appropriate grey level transformation is applied to enhance the darker regions of the image.

Suitable Grey Level Transformation

The most suitable technique is Log Transformation.

The mathematical expression is:

$$\boxed{s = c\log(1 + r)}$$

where:

- **r** = Original pixel intensity
- **s** = Transformed pixel intensity
- **c** = Constant

Concept and Working

Log transformation is a non-linear transformation that increases the intensity of low-value pixels more than high-value pixels. It expands the darker regions of the image while compressing brighter regions. This helps reveal hidden details without making the image excessively bright.

Justification

Log transformation is suitable because it enhances low-intensity regions where important information is present. It is widely used in medical images, satellite images, and X-ray images to improve visibility and make image analysis easier.

Expected Outcome

After applying log transformation, the dark regions become brighter, hidden details become visible, and the overall image quality and contrast are improved.

Conclusion

Therefore, **Log Transformation** is the most suitable grey level transformation for enhancing dark images because it improves low-intensity pixels and provides better visibility for further image analysis.

3.An image has poor contrast.
Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.
Introduction

An image has poor contrast when the pixel intensity values are concentrated within a narrow range. As a result, the difference between the object and the background becomes very small, making the image appear dull and reducing the visibility of important details.

Suitable Histogram-Based Technique

The most suitable technique is Histogram Equalization.

Concept and Working

Histogram Equalization is an image enhancement technique that redistributes the pixel intensity values over the entire available intensity range. It calculates the histogram of the image, determines the cumulative distribution function (CDF), and maps the original intensity values to new values. This spreads the pixel values more uniformly and increases the contrast of the image.

Justification

Histogram Equalization is suitable because it automatically enhances the contrast without changing the image content. It makes objects more distinguishable from the background and improves the visibility of edges and fine details. This technique is widely used in medical imaging, satellite imaging, and computer vision applications.

Expected Outcome

After applying Histogram Equalization, the image becomes clearer with improved contrast. Hidden details become visible, object boundaries are better defined, and the overall image quality is enhanced.

Conclusion

Therefore, **Histogram Equalization** is the most suitable histogram-based technique for improving poor-contrast images because it redistributes intensity values uniformly, resulting in better visibility and enhanced image quality.

4A low-contrast image requires uniform intensity distribution.
Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

Answer

Introduction

A low-contrast image has a narrow range of pixel intensity values, making objects difficult to distinguish from the background. To improve visibility, the intensity values must be distributed more uniformly.

Suitable Method

The most suitable method is **Histogram Equalization**.

Concept and Working

Histogram Equalization redistributes the pixel intensity values across the entire grayscale range. It computes the image histogram, calculates the cumulative distribution function (CDF), and maps the original intensity values to new intensity levels. This produces a more uniform distribution of pixel values.

Justification

Histogram Equalization increases the contrast of the image by utilizing the full intensity range. It improves the visibility of objects and fine details without changing the image content.

Expected Outcome

The enhanced image has a balanced intensity distribution, improved contrast, and clearer object boundaries, making it suitable for further image analysis.

Conclusion

Therefore, Histogram Equalization is the most appropriate method for improving low-contrast images because it redistributes intensity values uniformly and enhances image quality.

5.An image shows uneven brightness across regions.

Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

Answer

Introduction

Uneven brightness or non-uniform illumination occurs when different regions of an image have different lighting conditions. Some areas appear very bright, while others appear dark, making it difficult to observe important features and reducing the overall image quality.

Suitable Enhancement Technique

The most appropriate technique is **Contrast Limited Adaptive Histogram Equalization (CLAHE)**.

Concept

CLAHE is an advanced version of Histogram Equalization that improves the contrast of an image locally rather than globally. Instead of processing the entire image at once, it divides the image into small regions (tiles) and enhances each region separately. It also limits the contrast enhancement to avoid excessive noise amplification.

Working Process

1. Divide the image into small blocks or tiles.
2. Apply Histogram Equalization to each tile independently.
3. Limit the contrast using a clip limit to prevent over-enhancement.
4. Combine all enhanced tiles using interpolation to obtain a smooth and natural-looking image.

Justification

CLAHE is more effective than standard Histogram Equalization because it improves local contrast without creating excessive brightness or amplifying noise. It is widely used in medical imaging, satellite imaging, and microscopic image analysis where images often suffer from uneven illumination. **Expected Outcome**

After applying CLAHE, the brightness becomes more uniform across the image. Dark regions become clearer, bright regions remain balanced, local details are enhanced, and the overall image quality is significantly improved for further analysis.

6.A noisy image contains random intensity variations.

Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

Answer

Scenario Interpretation

A noisy image contains random intensity variations caused by sensor errors, electronic interference, or image transmission. These unwanted pixels reduce image quality and make feature extraction difficult. Therefore, a smoothing filter is applied before further image processing.

Suitable Smoothing Filter

The most suitable smoothing filter is the **Median Filter** because it effectively removes random impulse (salt-and-pepper) noise while preserving image edges.

Working Mechanism

The Median Filter uses a **3 × 3** or **5 × 5** neighborhood around each pixel.

Consider the following **3 × 3** pixel matrix:

$$\begin{bmatrix} 25 & 30 & 28 \\ 29 & 255 & 31 \\ 27 & 32 & 26 \end{bmatrix}$$

Here, **255** is a noisy pixel.

Arrange all the values in ascending order:

$$25, 26, 27, 28, 29, 30, 31, 32, 255$$

The **median value = 29**.

The center pixel (255) is replaced with **29**.

Filtered Matrix:

$$\begin{bmatrix} 25 & 30 & 28 \\ 29 & 29 & 31 \\ 27 & 32 & 26 \end{bmatrix}$$

Thus, the noisy pixel is removed while preserving the surrounding image information.

Justification

The Median Filter is preferred because:

- It removes salt-and-pepper noise effectively.
- It preserves image edges better than the Mean Filter.
- It does not blur object boundaries.
- It is widely used in medical imaging, document processing, and computer vision applications.

Impact on Noise Reduction

After applying the Median Filter:

- Random noisy pixels are removed.
- The image becomes smoother.
- Important edges and fine details are preserved.
- The processed image is suitable for segmentation and feature extraction.

7.**An image requires noise reduction without significant loss of details.**
Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

Answer

Scenario Interpretation

The image contains noise that affects its quality, but it is also important to preserve edges and fine details. If a normal smoothing filter is used, the noise may be reduced, but important features such as object boundaries can become blurred. Therefore, an edge-preserving filter is required.

Suitable Filtering Approach

The most suitable filtering approach is the **Bilateral Filter**.

Concept

The Bilateral Filter is a non-linear smoothing filter that reduces noise while preserving edges. It considers both the **spatial distance** between neighboring pixels and the **difference in their intensity values**. Pixels with similar intensity are averaged, while pixels with large intensity differences (edges) are preserved.

Working Process

Consider the following **3 × 3** image matrix:

$$\begin{matrix} 48 & 50 & 49 \\ [51 & 100 & 52] \\ 49 & 50 & 48 \end{matrix}$$

The center pixel **100** is affected by noise.

The Bilateral Filter compares both the position and intensity of neighboring pixels. Since the neighboring values are close to **50**, the noisy value **100** is replaced by a value close to the surrounding pixels while preserving edge information.

Filtered Matrix:

$$\begin{matrix} 48 & 50 & 49 \\ [51 & 50 & 52] \\ 49 & 50 & 48 \end{matrix}$$

8.An image appears blurred after smoothing.

Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

Scenario Interpretation

After applying a smoothing filter, the image noise is reduced, but the edges and fine details become blurred. This makes object boundaries unclear and reduces the quality of the image. Therefore, a sharpening technique is required to restore the lost details.

Suitable Sharpening Method

The most suitable sharpening method is the **Laplacian Filter**.

Concept

The Laplacian Filter is a second-order derivative operator that detects rapid changes in pixel intensity. It highlights image edges by emphasizing high-frequency components, making the image appear sharper.

Working Principle

The Laplacian Filter calculates the difference between a pixel and its neighboring pixels. Regions where the intensity changes rapidly are identified as edges. These edge values are then added back to the original image, resulting in a sharper image with clearer object boundaries.

Justification

The Laplacian Filter is preferred because it restores edge information lost during smoothing. It enhances fine details without changing the overall image structure and is widely used in medical imaging, satellite imaging, and object detection.

Expected Outcome

After applying the Laplacian Filter:

- Blurred edges become sharper.
- Fine details are restored.
- Object boundaries become more distinct.
- The image quality improves for further processing.

Conclusion

Therefore, the **Laplacian Filter** is an effective sharpening technique for restoring image details after smoothing. It enhances edges and improves the clarity of the image, making it suitable for feature extraction and analysis.

9.Fine details in an image are not clearly visible.

Describe and justify the filtering technique used to enhance details, explaining its effect on highfrequency components.

Answer

Scenario Interpretation

Fine details such as edges, textures, and object boundaries are not clearly visible because the high-frequency components of the image are weak or suppressed. This makes the image appear dull and reduces the visibility of important features. Therefore, a suitable filtering technique is required to enhance these details.

Suitable Filtering Technique

The most suitable filtering technique is the **High-Pass Filter (HPF).**

Concept

A High-Pass Filter is used to enhance the high-frequency components of an image, which represent edges, fine details, and sharp intensity changes. It suppresses the low-frequency components (smooth regions) while allowing the high-frequency information to pass through.

Working Principle

The High-Pass Filter detects regions where the pixel intensity changes rapidly and enhances those regions. As a result, edges become sharper, textures become more prominent, and fine image details are highlighted. It is commonly applied after smoothing to restore the sharpness of the image.

Justification

The High-Pass Filter is preferred because it improves edge visibility without affecting the overall structure of the image. It is widely used in image enhancement, feature extraction, medical imaging, satellite image processing, and object detection where fine details are important.

Effect on High-Frequency Components

After applying the High-Pass Filter:

- High-frequency components are enhanced.
- Edges become sharper.
- Fine details and textures become more visible.
- The image appears clearer and more suitable for analysis.

10.A noisy image must be preprocessed before further analysis.

Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

Answer

Scenario Interpretation

A noisy image contains unwanted pixel variations that reduce image quality and affect the accuracy of image processing tasks such as segmentation, feature extraction, and classification. Therefore, preprocessing is required to remove noise before further analysis.

Suitable Spatial Filtering Method

The most suitable spatial filtering method is the **Gaussian Filter**.

Concept

The Gaussian Filter is a linear smoothing filter that reduces Gaussian and random noise by applying a Gaussian-weighted kernel to the image. It smooths the image while preserving the overall image structure.

Working Principle

The Gaussian Filter uses a Gaussian kernel to calculate the weighted average of neighboring pixels. Pixels closer to the center are given higher weights than pixels farther away. This reduces random noise and smooths the image without causing excessive distortion.

Application

1. Read the noisy input image.
2. Select a Gaussian kernel (e.g., **3 × 3** or **5 × 5**).
3. Apply the Gaussian Filter to the image.
4. Replace each pixel with the weighted average of its neighboring pixels.
5. Obtain a smoother image with reduced noise.

Justification

The Gaussian Filter is widely used as a preprocessing technique because it effectively reduces random noise while maintaining the overall appearance of the image. It prepares the image for accurate segmentation, feature extraction, and object recognition. It is commonly used in medical imaging, computer vision, and image analysis applications.

Expected Outcome

After applying the Gaussian Filter:

- Random noise is reduced.

- The image becomes smoother and clearer.
- Image quality is improved.
- The processed image becomes suitable for further analysis such as segmentation and classification.